Voice2Care: AI-Powered Speech-to-Store&View Clinical Documentation

Final project - *Big Data Engineering, 2025 @ UNINA*

1 Introduction

The *Speech-to-Store&View* project aims to optimize clinical documentation through the implementation of advanced technologies, thereby reducing the manual workload for healthcare professionals. Currently, the manual transcription of clinical information is a time-consuming and error-prone process, particularly in high-intensity environments. A notable example is the emergency department, where a nurse onboard an ambulance must quickly record a patient's condition in a critical situation.

To address this challenge, a system can be designed to automatically convert audio recordings from physicians and/or nurses into text using advanced speech recognition models. However, in the medical domain, this technology still presents certain limitations[1] due to the complexity of medical terminology and the need for high accuracy. To enhance transcription reliability, the generated text can be further processed using natural language processing (NLP) algorithms, with a particular focus on the application of Large Language Models (LLMs). These models enable the systematic extraction and structuring of the most relevant clinical information, facilitating its interpretation and utilization.

The structured data can then be stored in a NoSQL database, ensuring fast access and efficient management of clinical information. Subsequently, an interactive dashboard can be implemented to allow healthcare professionals to visualize, edit, and validate the extracted data before its integration into the official medical records. Furthermore, the system may offer an additional feature that enables physicians to automatically generate structured patient reports in PDF format directly from the dashboard, further streamlining the documentation process and enhancing information traceability.

Adopting such an approach would not only significantly reduce transcription time but also improve the accuracy, usability, and accessibility of clinical data. Consequently, healthcare professionals would be able to make faster and more informed decisions, ultimately enhancing patient care quality and promoting the effective use of data-driven technologies in the medical sector.

[1] `https://bit.ly/3FIIVrU`

2 Methodology

The project follows a structured process divided into main phases (an example of workflow in Figure 1), with simplified guidelines to ensure a clear and effective implementation.

- **Speech-to-Text**: The first phase involves converting audio recordings into text. This step is performed using advanced speech recognition models or cloud services, such as Whisper AI[2].
- **Natural Language Processing (NLP) and Large Language Models (LLM)**: Once the text has been transcribed, NLP techniques and LLMs are applied to extract and structure the most relevant clinical information. For this phase, libraries from Hugging Face[3] can be utilized, including models like Gemma, Qwen, LLama, and others. The final output should be a structured JSON file containing fields consistent with those shown in Figure 2 (choosing a sufficient number of fields to make the view as representative as possible). The generation of the JSON can be facilitated using LLM-to-JSON tools, such as those listed here[4] or following this guide[5].
- **Storage in NoSQL Database**: The structured data in JSON format is stored in a NoSQL database, such as MongoDB, to ensure rapid access to information and system scalability.
- **Interactive Dashboard and Report Generation - Template Population**: The final phase involves developing an intuitive interface to visualize and manage the data. The dashboard can be implemented using frameworks such as Dash[6], Streamlit[7], or Shiny[8]. This interface should allow healthcare professionals to review, edit, and validate the processed information. Additionally, a module for generating structured PDF reports can be integrated, using libraries like ReportLab[9]. For data visualization, interactive charts available in Echarts[10] can be utilized.

Based on the methodology provided as a starting point, we ask you to explore the possibility of adding other techniques or modules as you wish, while maintaining the context and ensuring a certain coherence of the project.

[2] `https://github.com/openai/whisper`
[3] `https://huggingface.co/models`
[4] `https://github.com/imaurer/awesome-llm-json`
[5] `https://bit.ly/3FRUwVS`
[6] `https://dash.plotly.com/introduction`
[7] `https://docs.streamlit.io/`
[8] `https://shiny.posit.co/py/`
[9] `https://www.reportlab.com/dev/open-source-reportlab/`
[10] `https://echarts.apache.org/examples/en/index.html`

3 Objective

The main objective of the project is to optimize clinical documentation by ensuring that both the storage and the visualization of clinical data occur immediately. This rapid processing is crucial in time-sensitive environments, such as emergency departments, where decisions must be made swiftly and accurately.

3.1 Practical Examples

- **Example 1: Routine Checkup with Automatic Report Generation**
  During a routine checkup for a patient with diabetes, the doctor records observations using an audio device. The system immediately transcribes the recording, extracts key clinical terms (e.g., "blood sugar levels", "therapy adjustment"), and structures this information into a JSON report. The doctor can review, modify, and validate the report on the interactive dashboard, which instantly updates the patient's medical record.

- **Example 2: Emergency Department Scenario**
  In an emergency setting, a doctor records a voice note for a patient experiencing chest pain. The system rapidly transcribes the note, highlights critical keywords (e.g., "retro-sternal pain", "ECG required", "monitoring needed"), and displays the information on the dashboard. This enables the medical team to immediately verify the data and generate a PDF report for use in the emergency management process.

Create a dashboard that encapsulates the entire system in the most intuitive way for the user.

3.2 Managed Data

The system processes and manages the following types of data:

- *Audio/Voice Recordings* containing medical notes or consultations (can search for them online freely and/or generate them yourself).
- *Transcribed Text* obtained from speech recognition.
- *Structured JSON Data* containing clinically relevant information extracted from the transcribed text following various fields presented in Figure 2.
- *Patient History*, progressively updated over time.

An example of a dashboard and/or report structure is shown in this Figure 2 or in extension of Fascicolo Sanitario Elettronico[11]*. N.B. It is not necessary to extract all possible fields from the Figure*

[11] `https://www.fascicolosanitario.gov.it/cosa-contiene.html`

4 Tools and Technologies

To implement the project, the following tools and technologies are employed:

- **Speech Recognition Tools**: Software and cloud services for converting audio to text.
- **NLP and LLM Libraries**: Platforms like Hugging Face for analyzing and structuring the data.
- **Database Management**: MongoDB for scalable and efficient storage.
- **Interactive Dashboard**: Frameworks such as Dash, Streamlit, or Shiny for Python to visualize and modify data.
- **PDF Report Generation** (optional): Libraries such as ReportLab to create structured documents.

5 Project Evaluation

The project will be evaluated by a committee, which will assign a score from 0 to 5 to various aspects of the methodology and final dashboard. The evaluation of the methodology may vary based on the oral discussion and will be communicated only after the discussion.

5.1 Dashboard Evaluation

- **Usability**: The interface should be intuitive for healthcare professionals.
- **Accuracy**: The displayed information must be precise and clinically relevant.
- **Efficiency**: The system should allow for quick modifications and validations of documents.
- **Customization**: Users should be able to filter and adapt the displayed data according to their needs.
- **Dashboard/Report Generation**: The PDF generation option should produce well-structured and readable patient records.

5.2 Methodology Evaluation

- **Speech Recognition Accuracy**: The accuracy of the transcription compared to the original recordings.
- **Information Extraction Reliability**: The effectiveness of the NLP and LLM models in structuring clinical data.

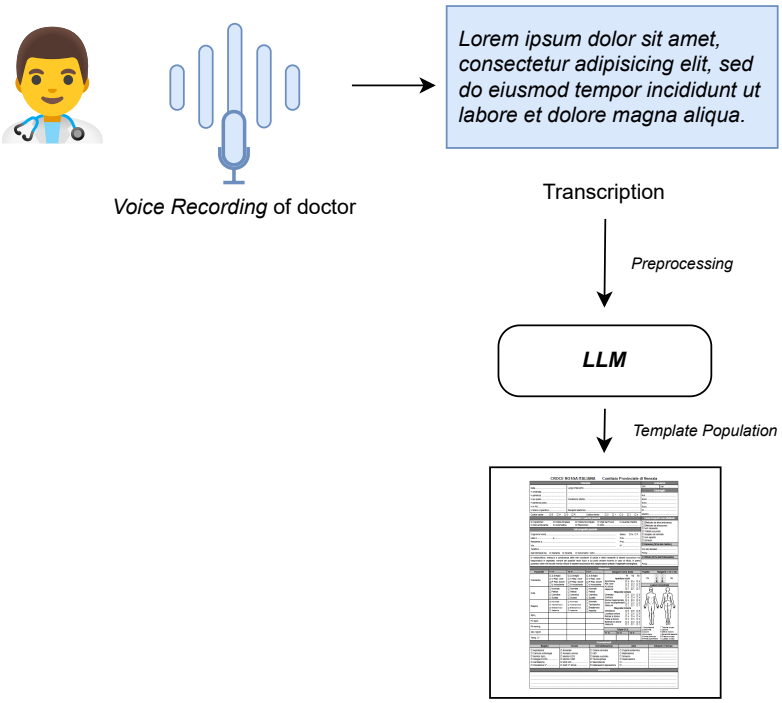

Figure 1: An example of Workflow

- **Scalability**: The ability of the system to handle large volumes of clinical data.
- **Security and Compliance**: Adherence to healthcare regulations and data protection standards.

CROCE ROSSA ITALIANA Comitato Provinciale di Venezia

Chiamata

Data..... Luogo intervento.....
H chiamata ....
H partenza..... .....
H sul posto ..... Condizione riferita .....
H partenza posto ....
H in PS ....
H libero e operativo..... Recapito telefonico.....

Codice uscita ☐ B ☐ V ☐ G ☐ R Codice rientro ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4

Ambulanza

CRI ..... Sel. .....

Equipaggio

Aut. .....
Socc.....
Socc.....
Socc.....
IP .....
Medico .....

Attivazioni / Autorità presenti

☐ Carabinieri ☐ Polizia Stradale ☐ Polizia Municipale ☐ Vigili del Fuoco ☐ Guardia medica
☐ Altra ambulanza ☐ Automedica ☐ Elisoccorso ☐ Altro.....

Causa trasporto non effettuato

☐ Effettuato da altra ambulanza
☐ Effettuato da elisoccorso
☐ Non necessita
☐ Trattato sul posto
☐ Sospeso da centrale
☐ Non reperito
☐ Scherzo

Dati anagrafici paziente

Cognome Nome ..... Sesso ☐ M ☐ F
Nato il.....a..... Prov .....
Residente a ..... Prov .....
Via..... N°.....
Telefono .....
Dati dichiarati da: ☐ Paziente ☐ Parente ☐ Documento / Altro .....

☐ Decesso (firma del medico)

Ora del decesso:.....
Firma .....

Io sottoscritto/a, messo/a a conoscenza delle mie condizioni di salute e della necessità di essere soccorso/a e/o trasportato/a in ospedale, nonché dei possibili rischi futuri a cui potrò andare incontro in caso di rifiuto, in pieno possesso delle mie facoltà mentali ***rifiuto di essere soccorso/a e/o trasportato/a presso l'ospedale consigliato.***

☐ Rifiuto (firma dell'interessato)

Firma .....

Rilevazioni

Parametri	T1 H:	T2 H:	T3 H: .....
Coscienza	☐ **A** Sveglio ☐ **V** Risp. voce ☐ **P** Risp. dolore ☐ **U** Incosciente	☐ **A** Sveglio ☐ **V** Risp. voce ☐ **P** Risp. dolore ☐ **U** Incosciente	☐ **A** Sveglio ☐ **V** Risp. voce ☐ **P** Risp. dolore ☐ **U** Incosciente
Cute	☐ Normale ☐ Pallida ☐ Cianotica ☐ Sudata	☐ Normale ☐ Pallida ☐ Cianotica ☐ Sudata	☐ Normale ☐ Pallida ☐ Cianotica ☐ Sudata
Respiro	☐ Normale ☐ Tachipnoico ☐ Bradipnoico ☐ Assente	☐ Normale ☐ Tachipnoico ☐ Bradipnoico ☐ Assente	☐ Normale ☐ Tachipnoico ☐ Bradipnoico ☐ Assente
SpO_2			
FC bpm			
PA mmHg			
Glic. Mg/dl			
Temp. C°			

Glasgow Coma Scale

	T1	T2	T3
Apertura occhi			
Spontanea	☐ 4	☐ 4	☐ 4
Alla voce	☐ 3	☐ 3	☐ 3
Al dolore	☐ 2	☐ 2	☐ 2
Nessuna	☐ 1	☐ 1	☐ 1
Risposta verbale			
Orientata	☐ 5	☐ 5	☐ 5
Confusa	☐ 4	☐ 4	☐ 4
Parole inappropriate	☐ 3	☐ 3	☐ 3
Suoni incomprensibili	☐ 2	☐ 2	☐ 2
Nessuna	☐ 1	☐ 1	☐ 1
Risposta motoria			
Obbedisce	☐ 6	☐ 6	☐ 6
Localizza dolore	☐ 5	☐ 5	☐ 5
Retrae al dolore	☐ 4	☐ 4	☐ 4
Flette al dolore	☐ 3	☐ 3	☐ 3
Estende al dolore	☐ 2	☐ 2	☐ 2
Nessuna	☐ 1	☐ 1	☐ 1

Totale GCS		
T1 H:.....	T2 H:	T3 H: .....

Pupille: **Reagenti ☐ Si ☐ No**

Dx Sx

Lesioni riscontrate

1 Amputazione
2 Deformità
3 Dolore
4 Emorragia
5 Ferita profonda
6 Ferita superficiale
7 Trauma chiuso
8 Ustione
9 Deficit motorio
A Sensibilità assente
B Frattura o sosp.
C Lussaz. o sosp.

Provvedimenti

Respiro	Circolo	Immobilizzazione	Altro	Infusioni / Farmaci
☐ Aspirazione ☐ Cannula orofaringea ☐ Monitor SpO_2 ☐ Ossigeno lt/min..... ☐ Ventilazione ☐ Intubazione N°.....	☐ Emostasi ☐ Accesso venoso ☐ Monitor ECG ☐ Monitor NIBP ☐ MCE min..... ☐ DAE N° Shock	☐ Collare cervicale ☐ KED ☐ Barella cucchiaio ☐ Tavola spinale ☐ Steccobenda ☐ Materassino depressione	☐ Coperta isotermica ☐ Medicazione ☐ Ghiaccio ☐ Osservazione ☐ ..... ☐	....

Annotazioni

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Figure 2: An example of Patient's Report Structure